

Practice Questions

1. Write a shell script to sort contents of a file and save it to another file using output redirection.

```
#!/bin/bash
echo "Enter input file:"
read infile
echo "Enter output file:"
read outfile
sort "$infile" > "$outfile"
echo "Sorted content saved to $outfile"
```

2. Write a shell script to append file content to another file using redirection.

```
#!/bin/bash
echo "Enter source file:"
read file1
echo "Enter destination file:"
read file2
cat "$file1" >> "$file2"
```

3. Write a shell command to display the **top 3 most frequent words** in a file using multiple pipes.

```
cat file.txt | tr ' ' '\n' | sort | uniq -c | sort -nr | head -3
```

Note:

- **cat file.txt** ☐ Displays the contents of the file.
- **tr ' ' '\n'** ☐ tr = translate characters
Replaces **spaces (' ') with newline ('\n')**
This converts text into **one word per line**
- **uniq -c** ☐ Counts **duplicate occurrences**
Works only on sorted data

- **sort -nr**

-n → numeric sort

-r → reverse order (descending)

4. Write a shell command to display **only duplicate words** in a file using multiple pipes.

```
cat file.txt | tr ' ' '\n' | sort | uniq -d
```

5. Write a shell command to count how many **unique words** are present in a file.

```
cat file.txt | tr ' ' '\n' | sort | uniq | wc -l
```

6. Write a shell command to display lines containing numbers, sorted and without duplicates.

```
cat file.txt | grep '[0-9]' | sort | uniq
```

7. Write a shell command to count how many times each word appears in a file.

```
cat file.txt | tr ' ' '\n' | sort | uniq -c
```

Sample Questions to try out:

8. Program to check whether a student is vaccinated or not and update the status in status.txt file.

```
# read vaccination status from student
# if not vaccinated - not allow to write exams
# first dose only - need rtPCR
# two doses - allow
# write the result to a file
```

```
# To check the number of arguments
if [ $# -lt 1 ]; then
    echo "Atleast 1 arguments needed";
    echo "0 - Not vaccinated"
    echo "1 - First dose only"
    echo "2 - Two doses"
    exit;
fi

if [ $1 -eq 0 ]; then
    echo "NOT ALLOWED" > status.txt;
elif [ $1 -eq 1 ]; then
    echo "Need to submit RTPCR " > status.txt;
elif [ $1 -eq 2 ]; then
    echo "Allow" > status.txt;
else
    echo "Invalid option";
fi

echo "give this command to show status : cat status.txt"
```